

**UDAYAN
GANGULY**
Professor-in-
Charge, SemiX,
IIT Bombay

MANUFACTURING CHIP SOVEREIGNTY

Udayan Ganguly of IIT Bombay says
it's a long, patient game

Jayesh Shinde | editor@digit.in

At the centre of India's semiconductor manufacturing push stands Udayan Ganguly – professor of electrical engineering at Indian Institute of Technology Bombay, a key architect of its nanofabrication ecosystem, and one of the country's most consistent voices arguing that chip design prowess alone won't secure technological sovereignty.

Ganguly returned to India in 2010 with something more ambitious than a research agenda: he wanted to help build manufacturing muscle. Over the years, his team has pioneered India's first indigenous CMOS-180nm 8-bit OTP memory technology – adopted by Semi-Conductor Labs (SCL), Chandigarh – and more recently developed a 64kbit PUF chip that earned recognition at the VLSI Design Conference and is undergoing military grade testing. Needless to say, his credentials are formidable.

Today, Ganguly serves as the first Professor-in-Charge of SemiX – IIT

Bombay's Centre for Semiconductor Technologies – a platform that brings together industry affiliates, national mission objectives, and advanced chip R&D under one roof. SemiX isn't just another academic centre, but a culmination of nearly two decades of infrastructure building, industry collaboration, and talent pipeline creation at IITB's Nanofab.

When I ask Ganguly what he means by IIT Bombay's "specialty" being manufacturing, he doesn't romanticize it. "IITB Nanofab is a lab where advanced manufacturing technologies and devices are invented and demonstrated at a prototype level for next-generation solutions," he says. "Powered by 45+ faculty members, 13 domains are covered, where each domain has examples of key solutions developed."

To understand what that really means, you have to zoom in – way in.

"Semiconductor chip manufacturing takes 1000-2000 steps done on 100+ pieces of equipment in an

integrated facility," Ganguly explains. "Our lab has 100+ equipment under the same roof to develop & innovate to solve key chip challenges."

Each machine has a "Live Equipment Status with 90+% uptime." This is not an academic sandbox where tools gather dust between semesters. It's a living organism. IITB Nanofab – modeled after Cornell and Stanford's – runs like a national asset.

And the cleanroom? That was phase one. "Cleanrooms are necessary because a dust particle (0.5 micrometre) is 20x bigger than transistors (0.025 micrometer or 25 nanometre)," he says.

IIT Bombay built a Class 1000 (ISO 6) cleanroom first, before layering in equipment and process capability. Funding came from the Ministry of Electronics, with leadership from then Principal Scientific Advisor Dr. R. Chidambaram. Applied Materials donated a \$9 million R&D toolset. The scale-up from 2006 to 2012 wasn't accidental – it was engineered.